THE IMAGE OF A QUASI-COMPACT ALGEBRAIC SPACE UNDER A SURJECTIVE MORPHISM IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement begins with a slogan block: it is part of the counted statement span, verbatim, and its definition is covered by the reference/slogan/history preamble span.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Surjectivity of a morphism of algebraic spaces on the underlying point sets. Definition definition-surjective, used at proof lines 1334-1336. Makes |X| -> |Y| surjective.
- Quasi-compactness of an algebraic space tested on the underlying point set. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-space, used at proof lines 1336-1337. Allows the topological argument to conclude.
- The image of a quasi-compact topological space under a continuous map is quasi-compact. Topology, Lemma topology-lemma-image-quasi-compact, used at proof lines 1338-1340. Is the topological input for part (1).
- Surjectivity is stable under base change and quasi-compactness is tested after base change. Lemma lemma-base-change-surjective and Lemma lemma-characterize-representable-quasi-compact, used at proof lines 1341-1347. Provides the surjective base change used in part (2).

Source reference. Original definition 03ME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4746–4746. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03MH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4749–4749. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

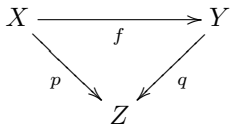
Source reference. Original lemma 03E4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4443–4443. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04Z9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6457–6457. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme.*

(1) *If $X \to Y$ is a surjective morphism of algebraic spaces over S, and X is quasi-compact then Y is quasi-compact.*
(2) *If*



is a commutative diagram of morphisms of algebraic spaces over S and f is surjective and p is quasi-compact, then q is quasi-compact.

Proof. Assume X is quasi-compact and $X \to Y$ is surjective. By Definition 03ME the map $|X| \to |Y|$ is surjective, hence we see Y is quasi-compact by Properties of Spaces, Lemma 03E4 and the topological fact that the image of a quasi-compact space under a continuous map is quasi-compact, see Topology, Lemma 04Z9. Let f, p, q be as in (2). Let $T \to Z$ be a morphism whose source is a quasi-compact algebraic space. By assumption $T \times_Z X$ is quasi-compact. By Lemma 03MH the morphism $T \times_Z X \to T \times_Z Y$ is surjective. Hence by part (1) we see $T \times_Z Y$ is quasi-compact too. Thus q is quasi-compact. □